

Functional Titration of Lentivirus (in K562 cells)

Introduction

This protocol is based on inducible CRISPRi/a K562 cells (suspension cells)
Using a 12-well plate (want 100 to 200 K cells per well and with a total volume of 1 mL)
Used RPMi with FBS and P/S
This protocol is written for one guide/virus; do these conditions per transduction
If doing adhesive cells, add a day -1 where you seed your cells. You will also have an easier time refreshing media (don't have to spin down)
Stock of the poly and puro should be in bacterial room -20C

Materials

- > 1.5 mL Eppendorf tubes
- > Normal media for your cell line (RPMI)
- > Polybrene
- > Puromycin
- > Cell Titer Glo reagents
- > Cell counter
- > Microplate reader
- > Cells
- > Virus
- >

Procedure

Day 0 (ex. Monday)

- Count cells and calculate total amount needed

For 9 wells, we spun down 3 M cells
No virus and no puro are necessary controls to calculate MOI

12-well plate layout (amount of virus)				
	A	B	C	D
1	10 uL	1 uL	10 ⁻¹	10 ⁻²
2	10 ⁻³	10 ⁻⁴	10 ⁻⁵	No puro
3	No virus			

- Resuspend in total media needed for all conditions

10 mL RPMI for this experiment

3. Want 5 ug/mL of polybrene

Stock = 1 mg/mL; added 50 uL of polybrene to our 10 mL

Be aware if you have any no poly controls

4. Add 1 mL of media/cells/poly mix to 1.5 mL falcon tubes

9 in this case

5. Add 10 uL of virus to your first tube, mix by pipetting with 1000 uL pipette 10x

6. Transfer 100 uL from tube 1 to tube 2, mix via pipetting with 100 uL pipette 10x

Use a new pipette between each tube! Previous dilution is at 10x concentration so a residual liquid in pipette can have a large effect

Continue till second to last tube

Leave the last tube alone

7. Add these tubes to their respective wells (1 mL in each)

8. Leave to grow in incubator

Day 2 (ex. Wednesday)

9. Create media containing 2 ug/mL of puromycin (want 2 ug per well)

Stock is 10 mg/mL --> create working dilution

1:10 --> add 10 uL stock puro to 90 uL media (RPMI)

10. Add 2uL of this puro/media to each tube, **except tube 8**

11. Gently pipette mix

Day 4 (ex. Friday)

12. Check out cells under microscope

Ideally cells are dead in no virus control

Idea are that 10 uL virus with cells are all alive, 1 uL are ~10% alive, 10⁻¹ are 1-2% alive and so forth

Try to get an idea of what this looks like visually

13. If cells are not dead in no virus control, consider refreshing media and leaving another day

14. Perform luminescence viability assay (Cell Titer Glo)

Use triplicates (or at least duplicates) for statistical/analytical purposes

Check cells under the microscope

15. Calculate MOI (want 0.2 to 0.4) and visualize results in R

Refer for MOI calculation doc

16. Continue culturing

Spin down cells and resuspend in puro/media mix (2 ug/mL)

Luminescence Viability Assay

17. Follow [CellTiter-Glo® Luminescent Cell Viability Assay Protocol](#)

Make sure cells equilibrate for 30 min at RT

Add 50 uL of Cell Titer glo buffer to each well

Pipette mix cells and add 50 uL of each condition/replicate to each well

Add to shaker for 2 minutes (500 rpm, 22C)

Incubate at RT for 10 minutes

18. Read on plate reader on alfa2

Days 6-14

19. Continue culturing cells that are withing reasonable MOI

consider keeping one condition above the reasonable MOI if cells are unhappy

20. Spin down cells and resuspend in puro/media mix (2 ug/mL)

21. Once cells are growing at a reasonable rate, transfer to a T75 with 11 mL of puro/media mix (2 ug/mL)